

JULES GOMEL

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PROFILE

Neuro-engineer and PhD candidate at ISAE-Supaero specializing in Brain–Computer Interfaces (BCI) and EEG signal processing. I bridge human-computer interaction and neuroscience to develop robust, user-centered neuromarkers. Passionate about open science and software development, I build reliable analytical pipelines to translate cognitive probing research into functional technology.

RESEARCH INTERESTS

Neural Entrainment and Visual Stimulation

EEG Signal Processing

Brain–Computer Interfaces (BCI)

RESEARCH EXPERIENCE

PhD Student

Dec 2024 – Present

ISAE-Supaero, supervised by Dr Frédéric and Dr Marie-Constance Corsi

Toulouse, France

- **PhD Topic:** Investigating EEG neuromarkers of visual information processing to probe cognition for diagnosis and BCI applications.
- **Completed Study:** Designed and conducted an EEG-based user study demonstrating that low-frequency textured flicker significantly improves neural entrainment and visual comfort for BCI control.
- **Ongoing Research:** Currently acquiring EEG data to investigate SSVEP and pupillometry modulation by textured flickers.
- Developed collaborative workflows using GitLab and organized a recurring journal club and research meetings for junior researchers.

Research Engineer

Mar 2024 – Nov 2024

ISAE-Supaero, supervised by Dr Frédéric Dehais

Toulouse, France

- Developed new features for the laboratory BCI platform including real-time visualization and performance evaluation using the Timeflux framework.
- Implemented dynamic feedback mechanisms for BCI experiments and conducted user studies evaluating their impact on performance and user experience.

Research Technician

Mar 2023 – Aug 2023

Drexel University, supervised by Dr Hasan Ayaz

Philadelphia, USA

- Evaluated Generative Adversarial Networks (GANs) for reconstruction of missing fNIRS signals and compared their performance with autoregressive models and interpolation techniques.

PUBLICATIONS

- **Gomel, J.**, Corsi, M.-C., & Dehais, F., Low-Frequency Textured Gabor Flicker Enhances Neural Entrainment and Visual Comfort for Brain-Computer Interface Control. (*under review*)
- **Gomel, J.**, Torre-Tresols, J., Cimarosto, P., Cabrera-Castillos, K., & Dehais, F., Effects of visual pre-decision feedback on user experience and decoding performance in cVEP-Burst BCI. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2025.
- **Merlau, B.**, et al. Assessing spatiotemporal and quality alterations in paretic upper limb movements after stroke in routine care: Proposal and validation of a protocol using IMUs versus MoCap. *Senors*, 2023.(*acknowledged*).

OPEN SOURCE PROJECTS

- **Study Repository & Dataset – Textured Gabor cVEP BCI:** Open-source repository containing the experimental design (PsychoPy) and EEG analysis pipeline (MNE-Python). The replication dataset is BIDS-compliant and deposited on Recherche Data Gouv. [[GitLab Repository](#)] & [[Dataset DOI](#)]
- **SSVEP Connectivity Toolbox** (*In development*): Python-based toolbox dedicated to the analysis of brain connectivity during Steady-State Visual Evoked Potentials (SSVEP). [[GitHub Repository](#)]

AWARDS

- Travel Award — 11th International BCI Meeting, Banff, Canada
- 2nd Place — Neuroergonomics 2024 Passive BCI Hackathon

CONFERENCES

- Cortico Days 2026 — Lille, France (Poster)
- IEEE Systems, Man and Cybernetics (SMC) 2025 — Vienna, Austria (Oral Presentation)
- BCI Meeting 2025 — Banff, Canada (Oral Presentation and Poster)
- Cortico Days 2025 — Lyon, France (Oral Presentation)

TECHNICAL SKILLS

Programming	Python (NumPy, MNE, Psychopy), C#
Signal Processing	EEG processing, time-frequency analysis, connectivity analysis
Machine Learning	Classification models, neural networks, GANs
BCI Frameworks	Timeflux
Experimental Methods	Human subject experiments, neuroergonomics protocols

TEACHING & MENTORING

- **Student Supervision:** Mentored a final-year engineering student project and a 2nd-year work-study student research project in neurosciences at ISAE-Supaero.
- **Lectures:** 6h on *Bridging Human Factors and Neurosciences with EEG Neuromarkers* for ISAE-Supaero master students.
- **Practical Sessions:** 30h on EEG signal processing, classification methods, and Brain–Computer Interface implementation for engineering students.

EDUCATION

Master of Neuroengineering and Signal Processing ISAE-Supaero, Toulouse, France Mobility in Politecnico di Milano, Italy and two internships in research during a gap year Thesis: <i>Development of a robust and comfortable cVEP BCI</i> Supervisor: Dr Frédéric Dehais	2021 – 2024
Bachelor of Mathematics, Physics and Engineering Science ISAE-Supaero, Toulouse, France	2018 – 2021

OUTREACH & ADDITIONAL ACTIVITIES

- **Science Communication:** Organized public talks on neuroergonomics and psychology during Neurodays 2024 in Toulouse (attendees : 50+).
- **Continuous Learning:** Completed advanced online certifications in *Medical Neurosciences* (Duke University).

- **Software Development:** Developed custom video game modifications using C#, applying advanced object-oriented programming and UX principles.
- **Strategy & Analysis:** Competitive Chess player.